

Hyperkalemia in Ambulant Postcardiac Surgery Patients during Combined Therapy with Angiotensin-converting Enzyme Inhibitor, Spironolactone, and Diet Rich in Potassium

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Published case report: PMC6489390 / PMID 30971598

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Case Report

Case no 1

36 year old female patient weighing 36 kgs, height 152 cm, admitted to our hospital with diagnosis of rheumatic heart disease, severe mitral stenosis (MS), severe aortic stenosis (AS), and severe tricuspid regurgitation (TR). The patient had dyspnea on exertion for the last 1½ years, with recent deterioration to dyspnea of New York Heart Association Class III. She was admitted with CHF 7 months back and managed conservatively. On examination, the patient had a pulse rate of 100 beats/min (bpm) in atrial fibrillation rhythm, mild pallor, and icterus, and pedal edema was present. Cardiovascular examination findings were suggestive of MS and AS. Hepatomegaly was present.

She was started on tablet furosemide 40 mg once daily (OD) along with tablet metoprolol 25 mg twice a day.

She was operated and mitral valve was replaced with size 27 St. Jude's mechanical valve and aortic valve with size 17 regent mechanical valve. Tricuspid commissurotomy was performed. Immediate postoperative period was uneventful and all predischage investigations were within normal range. She was discharged on the postoperative day (POD) 9 with prescription of tablet furosemide 40 mg OD, tablet aldactone 100 mg OD, tablet ramipril 5 mg HS, and tablet amiodarone 200 mg OD, and ecosprin 150 mg OD. On discharge, the patient did not have pedal edema or hepatomegaly.

On POD 28, her medication was changed to furosemide with spironolactone 40/25 mg BD, tablet acenocoumarol, tablet aldactone 100 mg OD, and tablet digoxin 0.25 mg OD. High dose of diuretics was used for a short while as the patient still had features of fluid overload. Postoperative echocardiography showed normal prosthetic valve function with normal biventricular function and moderate TR.

On POD 33, she presented in the emergency department with complaints of loose stool and vomiting (3 episodes) with severe muscular weakness. On examination, pulse was 36 bpm, irregular, systolic BP 60

mmHg, extremities were cold and clammy, signs of dehydration present, peripheral pulses not palpable, valve click present, and tender hepatomegaly present. Serum electrolytes showed hyperkalemia with serum potassium of 7.87 meq/L, hyponatremia with serum sodium 129 mmol/L and hypocalcemia, hemoglobin 9.6 gm% and hematocrit 38%, pH 7.28, base deficit-11, serum osmolality 252 mmol/L, serum creatinine 1.1 mg%, and blood urea nitrogen (BUN) 17 mg/dl. Electrocardiograph (ECG) showed heart rate of 30 beats/min, broad QRS complex, tented T-wave, and impending sine wave pattern. When detailed history was elicited from her husband, he revealed that apart from her medication, she was taking 7–8 bananas every day as part of her regular diet for the last 3 weeks.

She was managed with aggressive antihyperkalemic treatment with intravenous (IV) calcium gluconate, glucose-insulin drip, soda-bicarbonate, loop diuretics, K⁺ binding resins, and IV fluids. Over 24 h, she was stabilized in the intensive care unit (ICU) with same treatment. Hemodialysis was not required, she recovered hemodynamically and maintained good urine output. Potassium levels were serially monitored and were on a decreasing trend and finally levels decreased to 4.5 meq/L. ECG showed normal sinus rhythm, rate of 80–90 bpm. She was discharged from ICU after 2 days. She was advised to eat 1–2 bananas every day and the dose of potassium-sparing diuretics was reduced. She remained in a regular follow-up and no such incidence recurred.

Case no 2

8 years old child weighing 21 kgs, height 130 cm, admitted to our hospital with diagnosis of tetralogy of Fallot physiology for intracardiac repair. Examination showed a cyanotic child with pulse rate of 78 bpm in sinus rhythm and blood pressure of 100/70 mmHg. Preoperatively, the patient was continued on tablet propranolol 10 mg 8th hourly.

Perioperatively, dacron patch closure of ventricular septal defect, infundibular resection done, and transannular bovine pericardial patch placed. On POD 1, the patient developed complete heart block and was treated with temporary pacing, and injection dexamethasone was started, which was tapered off gradually and stopped on POD 10. Other medications included tablet furosemide 20 mg OD and tablet captopril 12.5 Mg 8th hourly. For nutrition, he was advised to include banana in the diet. Gradually over next week, the patient developed hepatomegaly and weight gain of 4 kg and was started on tablet aldactone 25 mg BD. The patient was then discharged on POD 15 after features of CHF were relieved.

On POD 21, he presented in the emergency department with nausea and vomiting, electrolytes showed K⁺ of 7.12 meq/L, Na⁺ 121 meq/L, serum creatinine 1.2 mg%, and BUN 15 mg%. ECG showed hyperkalemic changes. He was shifted to ICU and antihyperkalemic treatment started that included IV calcium gluconate, sodium bicarbonate, glucose-insulin drip, and potassium-binding resin. He was maintaining good hemodynamics with adequate urine output. Potassium levels were serially monitored and after 24 h of aggressive management, his K⁺ level was 4 meq/l. Dialysis was not required. He was discharged from ICU after 2 days. Retrospectively on interrogating further his father said that the boy was eating at least 6–7 bananas every day for the last 15 days. He was advised to eat a single banana every day and he remained in regular follow-up and had no such episode again.